

# Nikolai Daelemans

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## SUMMARY

Software Engineer & CTO with experience in AI automation, cloud infrastructure, and full-stack development. Currently building AvantifAI while leading AI platform engineering at KBC. Open to hybrid/remote engineering roles in AI, infrastructure, or technical leadership.

## SKILLS

**Tech Stack:** Python, Java, C#, JavaScript/TypeScript, React, Angular, Vue, SQL, Postgres  
**AI/ML:** PyTorch, TensorFlow, LLMs, RAG, Vector Databases, NLP, Prompt Engineering, ML Engineering  
**Sys/Infra:** Azure, AWS, Docker, Kubernetes, CI/CD, REST API, Microservices, DevOps, Kafka, Redis, Elastic, Jenkins, Git  
**Data & Analytics:** PowerBI, Fabric, Kibana, PySpark, Pandas  
**Other:** Agile/Scrum, Jira, Firewall/NAS/SAN/GSLB/Netscaler/F5, Zero Trust, IaC  
**Languages:** Dutch (C2), English (C2), French (B2), German(A2), Luxembourgish (A2)  
**Certifications:** Microsoft Certified: Azure AI Fundamentals (2025), Datacamp: Efficient AI model training with Pytorch, Introduction to PowerBI

## EXPERIENCE

### CTO & Co-founder — AvantifAI ([www.avantifai.com](http://www.avantifai.com))

*Based in Europe (March 2026 - present)*

- Running an AI consultancy designing automation tools for enterprise clients, reducing manual documentation overhead by hours per week.
- Developed DocAI — an intelligent enterprise documentation manager featuring agentic AI, multi-tenancy, Auth0 authentication, vector search, version control, and automated backups. Currently serving real pilots.

### Infrastructure Software Engineer (AI Platforms & Automation) — KBC

*Leuven, Belgium (July 2025 - present)*

- Built reusable **IaC frameworks** automating provisioning across 5+ infrastructure systems including compute, storage, firewall, load balancing and DNS, with AI-assisted decision making.
- Automated the process of users requests for Compute (On-Prem Virtualization), Storage operations (VMware, SAN, NAS, XP8), Firewall (Rules, Zero Trust, IDS, IPS), Load Balancing (L4, L7), DDI (DNS), Draining tool (Netscaler, F5)
- Team lead** of KBC's draining tool — organizing sprints and managing delivery in an agile environment using Jira.
- AI Innovation & Engineering** - Developing custom AI models and integrating Microsoft Copilot and Power Automate solutions to streamline internal banking processes. Troubleshooting recurring issues reported via internal support channels and automating resolutions to reduce human error.

### Technology Consultant — PwC

*Brussels, Belgium (February 2025 - May 2025)*

- Led the organization and technical preparation together with Microsoft of a hackathon for the NATO, coordinating multi-disciplinary teams and aligning deliverables with defense innovation goals, resulting in actionable MARCOM data solutions and strategic prototypes that informed NATO maritime safety protocols.
- Automated a time-intensive multi-excel workflow used by cross-functional teams, developing a Python script to ingest, process, and insert records, eliminating manual data entry and saving 20+ engineering hours per month.
- Preprocessed and cleaned large-scale defense datasets using Python (Pandas, NumPy), enabling accurate analytics and feature engineering.
- Developed dynamic visualizations and dashboards with Power BI and Plotly to highlight key patterns and threats from complex data structures, improving interpretability and decision-making for NATO analysts and leadership.

### Software Engineer — Tejo

*Hybrid (October 2024 - December 2024)*

- Designed and built a full event payment pipeline for a local non-profit — hosted frontend, Stripe/payment integration, automated confirmation emails with generated QR codes for event entry. Handled 100% digital payment processing with zero manual intervention on event day.

### Software Engineer (Product Owner) — Belfius

*Belgium (September 2024-December 2024)*

- Coordinated go-to-market technical readiness for an internal product modernization at Belfius, aligning engineering deliverables with business launch milestones and stakeholder requirements improving system performance by 50%.
- Designed and deployed scalable microservices using .NET Core and REST APIs, while contributing to Angular and React-based front-end enhancements to improve UX for internal users.

### Lead IT & Analytics — Emergent

*Leuven, Belgium (March 2024-March 2025)*

- Led the digital strategy and analytics initiatives for a student-run AI & Data Science organization, partnering with top-tier companies such as **McKinsey, KBC, AWS, Deloitte, DataCamp** and more to host workshops, talks, and hands-on projects; enhanced community engagement by 30% through scalable web solutions (WordPress/Elementor) and analytics-driven content planning (Python, PySpark).

### Software Engineer (App Development) — SunnySet

*Hybrid (January 2024 - September 2024)*

- Built and deployed a cross-platform mobile app using Flutter and Java to deliver real-time safety alerts and local news for tourists, enhancing urban navigation and user security in unfamiliar cities.
- Led core contributions across app logic, UI, and hosting, integrating CI/CD pipelines and managing web deployment to ensure smooth performance across devices.

### Software Engineer — Think-E

*Leuven, Belgium (July 2023 - September 2023)*

- Developed energy-efficient smart home applications by integrating IoT devices with real-time monitoring systems across residential complexes; leveraged C++, JavaScript, and SQL to connect sensors via Raspberry Pi, optimizing sustainability data collection and automation.
- Collaborated on full-stack solutions to improve energy tracking dashboards, incorporating responsive UI (HTML/MUI/CSS/JS) and GitHub-based version control to support scalable, maintainable systems.

## EDUCATION

**B.Sc. Applied Computer Science — UCLL, Belgium (2025)**

## PROJECTS

**2025 — AI & Infrastructure Learning Platform (*In Progress*):** Personal project building a learning platform for AI and infrastructure concepts. Hands-on experimentation with Raspberry Pi, IoT sensors, and custom model training. Focused on bridging theory and practical implementation.

**2024 — AI Questionnaire Generator and Interview Analysis System:** Built using Python, NLP, and LLMs to automate interview analysis.

**2023 — Campus Assistant Chatbot (<https://github.com/NikolaiDoesCode/CampusAI>)** Hybrid chatbot combining a custom-trained PyTorch intent classifier with DialoGPT fallback. Bag-of-words preprocessing pipeline, 5-layer MLP, deployed via Flask with a responsive frontend.

**2022 — FaceDetectorCNN (<https://github.com/NikolaiDoesCode/Face-Detection>):** Built a real-time face detection system from scratch using a custom CNN in PyTorch. Trained on 26,000 samples (LFW + generated negatives), achieving 99.89% validation accuracy and AUC-ROC of 1.0. Implemented sliding window detection, image pyramid, and non-maximum suppression from scratch.